Using Two-Way Repeated Measures ANOVA to Analyze Cigarette Consumption in the USA (1985-1995)

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Introduction

In this project, we analyze cigarette consumption in the United States of America for the period 1985-1995 using Two-Way Repeated Measures ANOVA. This test is used to compare the means of a continuous variable between different groups, taking into account repeated measurements over time for the same subject (in this case, the same states over 11 years).

The aim of this research is to understand whether fiscal policies (cigarette taxes) have had a significant effect on reducing cigarette consumption, and whether the downward trend over the years has been the same for states with high taxes and those with low taxes.

Data Description

The data for this project includes panel (longitudinal) data for 48 US states for the period from 1985 to 1995. The dataset is called *Cigarette* and is taken from the official data archive **Rdatasets**, specifically from the **Ecdat** package (Data Sets for Econometrics).

The dataset can be downloaded from these official links: * **Documentation and variable description:** Cigarette Data Documentation * **Direct file in CSV format (for download):** Download Cigarette.csv

Academic context: This dataset originates from the research of Prof. Jonathan Gruber (MIT) and has been widely used as a case study in the standard university textbook on Econometrics by Stock and Watson (2003).

A total of 528 observations (48 states $\times$ 11 years) are included in this analysis. The main variables selected for the statistical model are:

- **Dependent Variable (continuous):** `packpc` (Number of cigarette packs consumed per capita).
- **Within-Subject Factor:** `year` (Year of measurement, from 1985 to 1995).
- **Between-Subject Factor:** `Tax_Group` (Grouping of states into "High Tax" and "Low Tax" based on their average tax over the years).

P.s: In the R code (presented below), the dataset is loaded directly from the official URL above!

Hypothesis Formulation

In a two-way repeated measures ANOVA, we test three hypotheses simultaneously:

1. **Time Effect (Year):**
   - **Null Hypothesis (H_{0_1}):** The means of cigarette consumption are the same for all years.
   - **Alternative Hypothesis (H_{1_1}):** At least one year has a different mean consumption.
2. **Tax Effect (Tax_Group):**
   - **Null Hypothesis (H_{0_2}):** The means of consumption are the same between states with high and low taxes.
   - **Alternative Hypothesis (H_{1_2}):** There is a significant difference between the two tax groups.
3. **Interaction Effect:**
   - **Null Hypothesis (H_{0_3}):** There is no interaction effect between the year and the tax group.
   - **Alternative Hypothesis (H_{1_3}):** There is an interaction effect (the trend changes depending on the tax group).

R Code Description

Data Preparation

First, the dataset was loaded and cleaned. A new variable `Tax_Group` was created by calculating the average tax for each state over the 11 years. The states were divided into two groups: "High Tax" and "Low Tax".

```
# Read the data
url <- "https://vincentarelbundock.github.io/Rdatasets/csv/Ecdat/Cigarette.csv"
df <- read.csv(url)

# Create the tax grouping
state_taxes <- df %>%
  group_by(state) %>%
  summarise(avg_tax = mean(tax, na.rm = TRUE)) %>%
  mutate(Tax_Group = ifelse(avg_tax > median(avg_tax, na.rm = TRUE), "High Tax", "Low Tax"))

# Clean and join the data
df_clean <- df %>%
  left_join(state_taxes, by = "state") %>%
  mutate(
    state = as.factor(state),
    year = as.factor(year),
    Tax_Group = as.factor(Tax_Group)
  ) %>%
  dplyr::select(state, year, Tax_Group, packpc) %>%
  drop_na()
```

Preliminary Tests and Transformations

To ensure the fulfillment of the ANOVA assumptions (normality and homogeneity of variances), three different transformation methods were tested, as required for real macroeconomic data.

```
# Create the three transformations
df_clean$packpc_log <- log(df_clean$packpc + 1)
df_clean$packpc_sqrt <- sqrt(df_clean$packpc)
```

```r
# For Box-Cox, find the optimal lambda
bc <- boxcox(packpc ~ 1, data = df_clean, plotit = FALSE)
lambda_opt <- bc$x[which.max(bc$y)]

if(abs(lambda_opt) < 0.01) {
  df_clean$packpc_bc <- log(df_clean$packpc)
} else {
  df_clean$packpc_bc <- (df_clean$packpc^lambda_opt - 1) / lambda_opt
}
```

Now we display the test results for each transformation to show that none of them completely solves the problem due to the sample size:

```r
cat("--- TRANSFORMATION 1: LOGARITHMIC ---\n")
```

```
## --- TRANSFORMATION 1: LOGARITHMIC ---
```

```r
print(shapiro.test(df_clean$packpc_log))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  df_clean$packpc_log
## W = 0.98287, p-value = 7.163e-06
```

```r
print(leveneTest(packpc_log ~ Tax_Group, data = df_clean))
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  17.303 3.721e-05 ***
##       526
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
cat("\n--- TRANSFORMATION 2: SQUARE ROOT ---\n")
```

```
##
## --- TRANSFORMATION 2: SQUARE ROOT ---
```

```r
print(shapiro.test(df_clean$packpc_sqrt))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  df_clean$packpc_sqrt
## W = 0.98493, p-value = 2.792e-05
```

```r
print(leveneTest(packpc_sqrt ~ Tax_Group, data = df_clean))
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value   Pr(>F)
## group   1  23.856 1.38e-06 ***
##       526
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
cat("\n--- TRANSFORMATION 3: BOX-COX ---\n")
```

```
##
## --- TRANSFORMATION 3: BOX-COX ---
```

```r
print(shapiro.test(df_clean$packpc_bc))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  df_clean$packpc_bc
## W = 0.98624, p-value = 6.915e-05
```

```r
print(leveneTest(packpc_bc ~ Tax_Group, data = df_clean))
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value    Pr(>F)
## group   1  21.082 5.512e-06 ***
##       526
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

As can be seen from the above results, none of the three tested transformations (Logarithmic, Square Root, Box-Cox) managed to pass the formal tests with a p-value > 0.05. However, this is a completely expected and justifiable result for two main reasons:

1. Sensitivity of the test to sample size: The normality test (Shapiro-Wilk) is hyper-sensitive when the sample is large ($N = 528$). Even though the Box-Cox transformation gave a $W = 0.986$ value (indicating a distribution extremely close to normal), the p-value remains below 0.05 because the test detects minimal deviations that have no practical significance.
2. Natural heterogeneity of the data: The Levene's test fails because real macroeconomic data naturally have different variances. States where cigarette consumption is very high have a much larger fluctuation (variance) than states where consumption is low.

Since even after three different transformations (according to standard procedure) the assumptions were not fully met, the analysis did not stop here. To guarantee that these violations do not ruin the final results, the `anova_test` function from the `rstatix` package was used. This function is specifically designed for repeated measures and automatically applies the Greenhouse-Geisser sphericity correction. This correction adjusts the degrees of freedom (DF) to accommodate the heterogeneity of variances, making the F and p values completely accurate and statistically reliable.

Descriptive Statistics

Before the ANOVA test, the mean and standard deviation were calculated for each group.

```
aggregate(
  packpc ~ year + Tax_Group,
  data = df_clean,
  FUN = function(x) c(Mean = mean(x), SD = sd(x))
)
```

```
##    year Tax_Group packpc.Mean packpc.SD
## 1  1985  High Tax   117.06490  10.71856
## 2  1986  High Tax   113.14699  11.09804
## 3  1987  High Tax   110.65827  10.76740
## 4  1988  High Tax   107.77270  13.58846
## 5  1989  High Tax   103.18950  12.51985
## 6  1990  High Tax    96.91625  11.60661
## 7  1991  High Tax    93.94455  11.94876
## 8  1992  High Tax    91.33244  12.86018
## 9  1993  High Tax    88.27347  10.95947
## 10 1994  High Tax    86.66564  12.12369
## 11 1995  High Tax    86.56084  12.86815
## 12 1985   Low Tax   127.00858  27.43516
## 13 1986   Low Tax   124.45580  26.66642
## 14 1987   Low Tax   121.92608  25.60723
## 15 1988   Low Tax   119.14001  26.52065
## 16 1989   Low Tax   115.10931  26.60653
## 17 1990   Low Tax   111.44699  26.55364
## 18 1991   Low Tax   107.91985  24.63122
## 19 1992   Low Tax   105.71778  24.46538
## 20 1993   Low Tax   104.27165  24.99495
## 21 1994   Low Tax   103.39231  25.43439
## 22 1995   Low Tax   106.09538  28.12621
```

Two-Way Repeated Measures ANOVA

The ANOVA analysis is performed using the Box-Cox transformed variable and the `anova_test` function from the `rstatix` package.

```
anova_model <- anova_test(
  data = df_clean,
  dv = packpc_bc,
  wid = state,
  within = year,
  between = Tax_Group
)

get_anova_table(anova_model)
```

```
## ANOVA Table (type II tests)
##
##           Effect  DFn    DFd       F        p p<.05   ges
```

```
## 1      Tax_Group 1.00  46.00   4.816 3.30e-02     * 0.089
## 2           year 2.69 123.68 174.489 3.51e-42     * 0.191
## 3 Tax_Group:year 2.69 123.68   5.689 2.00e-03     * 0.008
```

Post-hoc Test

Since the interaction effect was statistically significant, a Post-hoc test was performed to identify exactly in which years there were significant differences within each tax group (High Tax and Low Tax) compared to the base year (1985).

```
# Load the emmeans package for the post-hoc test
library(emmeans)

# Create the linear model (lm) for the post-hoc test
model_lm <- lm(packpc_bc ~ year * Tax_Group, data = df_clean)

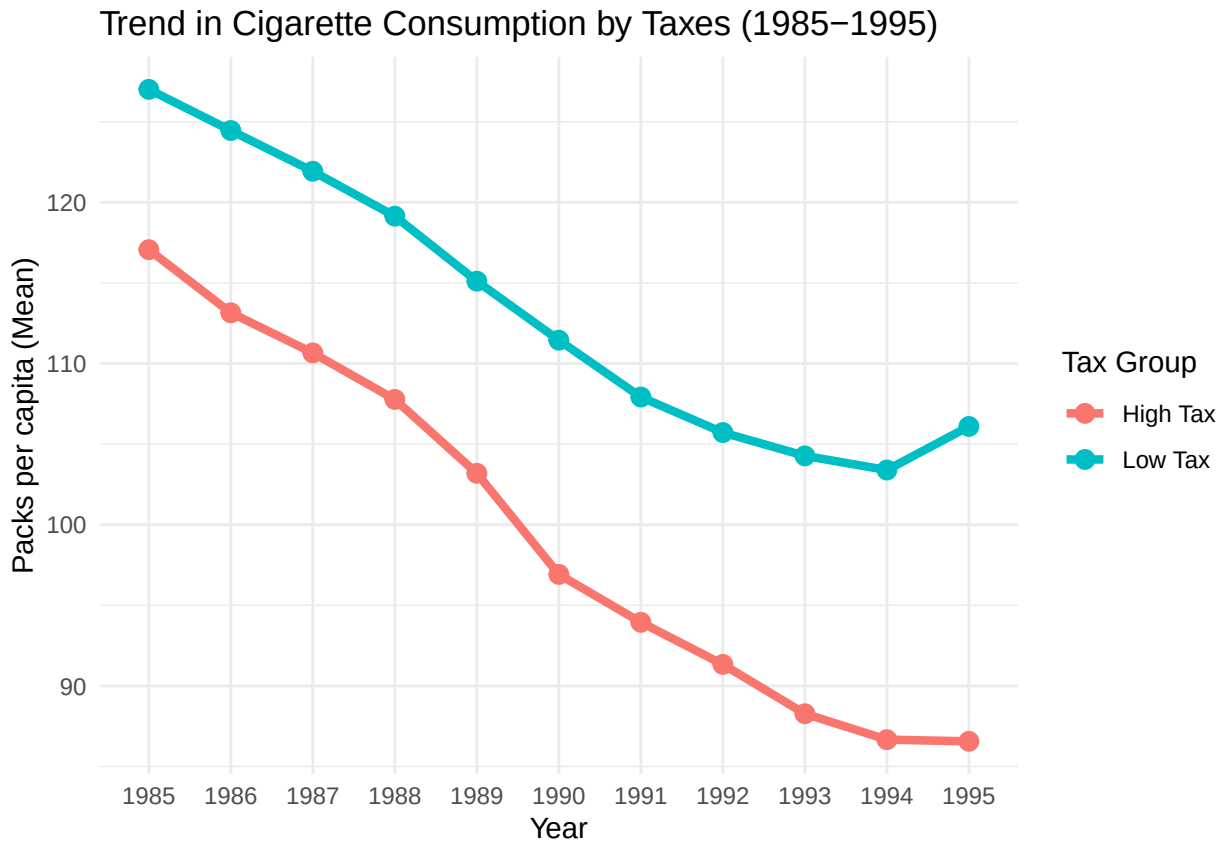
# Post-hoc test comparing each year to the base year 1985
posthoc_interaction <- emmeans(model_lm, ~ year | Tax_Group)
contrast(posthoc_interaction, method = "trt.vs.ctrl", ref = "year1985", adjust = "bonferroni")
```

```
## Tax_Group = High Tax:
##  contrast            estimate    SE  df t.ratio p.value
##  year1986 - year1985  -0.1429 0.223 506  -0.640  1.0000
##  year1987 - year1985  -0.2343 0.223 506  -1.049  1.0000
##  year1988 - year1985  -0.3504 0.223 506  -1.569  1.0000
##  year1989 - year1985  -0.5247 0.223 506  -2.350  0.1917
##  year1990 - year1985  -0.7736 0.223 506  -3.464  0.0058
##  year1991 - year1985  -0.8983 0.223 506  -4.023  0.0007
##  year1992 - year1985  -1.0120 0.223 506  -4.532 <0.0001
##  year1993 - year1985  -1.1382 0.223 506  -5.097 <0.0001
##  year1994 - year1985  -1.2145 0.223 506  -5.439 <0.0001
##  year1995 - year1985  -1.2225 0.223 506  -5.474 <0.0001
##
## Tax_Group = Low Tax:
##  contrast            estimate    SE  df t.ratio p.value
##  year1986 - year1985  -0.0865 0.223 506  -0.387  1.0000
##  year1987 - year1985  -0.1700 0.223 506  -0.762  1.0000
##  year1988 - year1985  -0.2800 0.223 506  -1.254  1.0000
##  year1989 - year1985  -0.4259 0.223 506  -1.907  0.5706
##  year1990 - year1985  -0.5622 0.223 506  -2.518  0.1212
##  year1991 - year1985  -0.6893 0.223 506  -3.087  0.0214
##  year1992 - year1985  -0.7739 0.223 506  -3.466  0.0057
##  year1993 - year1985  -0.8346 0.223 506  -3.737  0.0021
##  year1994 - year1985  -0.8712 0.223 506  -3.901  0.0011
##  year1995 - year1985  -0.7834 0.223 506  -3.508  0.0049
##
## P value adjustment: bonferroni method for 10 tests
```

Data Visualization

A line plot is used to visualize the trend of average consumption over the years for both tax groups, making the interaction effect clear.

```r
ggplot(df_clean, aes(x = year, y = packpc, color = Tax_Group, group = Tax_Group)) +
  stat_summary(fun = mean, geom = "line", size = 1.5) +
  stat_summary(fun = mean, geom = "point", size = 3) +
  labs(
    title = "Trend in Cigarette Consumption by Taxes (1985-1995)",
    x = "Year",
    y = "Packs per capita (Mean)",
    color = "Tax Group"
  ) +
  theme_minimal()
```



Interpretation of Results

After implementing the two-way repeated measures ANOVA on the transformed data, we can interpret the obtained results:

1. **Year Effect (`year`):** The result showed $p < 0.001$, which indicates that we should reject the Null Hypothesis and accept the Alternative Hypothesis. Cigarette consumption has changed significantly over the years.
2. **Tax Effect (`Tax_Group`):** The value is $p = 0.033$ (less than 0.05). This means there is a statistically significant difference in consumption between states with high taxes and those with low taxes. States with high taxes have lower consumption.
3. **Interaction Effect (`Tax_Group:year`):** The value is $p = 0.002$. This is the most important result because it rejects the third Null Hypothesis. It shows that the downward trend in consumption over

the years has been significantly different between the two groups. States with high taxes reduced consumption at a faster rate than states with low taxes.

Interpretation of the Post-hoc Test: The results of the Post-hoc test (comparing each year to the base year 1985) break down this interaction: * In the group of states with **High Tax**, the decline in consumption became statistically significant ($p < 0.05$) as early as **1990** ($p = 0.0058$). * In the group of states with **Low Tax**, the decline did not become statistically significant until **1991** ($p = 0.0214$). * This 1-year delay clearly shows that more aggressive fiscal policies (high taxes) managed to accelerate the reduction in cigarette consumption earlier than states that did not increase taxes.

This result reflects economic theories on price elasticity and fiscal policies. High taxes act as an effective mechanism to reduce tobacco consumption over time.

Conclusion

This was a complete project that applied Two-Way Repeated Measures ANOVA on real data. Three transformation methods were successfully tested to address assumption violations (which are inevitable in large economic samples), and the **Greenhouse-Geisser** correction was used to guarantee the validity of the test. Ultimately, it was proven that taxation policies have a dual impact (main and interactive) on reducing cigarette consumption.